Chapter 1 — Measurements

Board-style MCQs · PECTAA Physics 11 · Answer key at end

Practice MCQs

1. The number of base SI units is:

(a) 5 **(b)** 9 **(c)** 7 **(d)** 6

2. The SI unit of luminous intensity is:

(a) candela **(b)** ampere **(c)** mole **(d)** kelvin

3. The SI unit of amount of substance is:

(a) kelvin **(b)** mole **(c)** candela **(d)** kilogram

4. Which is a derived unit?

(a) newton **(b)** second **(c)** kelvin **(d)** metre

5. The SI unit of pressure is:

(a) newton **(b)** watt **(c)** joule **(d)** pascal

6. One nanometre equals:

(a) 10^{-12} m **(b)** 10^{-3} m **(c)** 10^{-9} m **(d)** 10^{-6} m

7. The prefix mega stands for:

(a) 10^{-6} **(b)** 10^{6} **(c)** 10^{3} **(d)** 10^{9}

8. 1 μF is equal to:

(a) 10^{-3} F **(b)** 10^{-6} F **(c)** 10^{-12} F **(d)** 10^{-9} F

9. Compound prefixes are:

(a) not allowed **(b)** allowed **(c)** required **(d)** used for base units

10. 1 cm³ equals:

(a) 10^{-3} m³ **(b)** 10^{-9} m³ **(c)** 10^{-2} m³ **(d)** 10^{-6} m³

11. The least count of a metre rule is usually:

(a) 0.1 cm **(b)** 0.001 cm **(c)** 0.01 cm **(d)** 1 cm

12. The least count of a vernier callipers is commonly:

(a) 0.001 cm **(b)** 0.01 cm **(c)** 1 cm **(d)** 0.1 cm

13. The least count of a micrometer screw gauge is commonly:

(a) 0.1 cm **(b)** 0.001 cm **(c)** 0.0001 cm **(d)** 0.01 cm

14. Uncertainty in a single reading is usually taken as:

(a) percentage error **(b)** zero **(c)** least count **(d)** mean value

15. For repeated readings, uncertainty is estimated by:

(a) maximum reading **(b)** mean deviation **(c)** least count only **(d)** zero error

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16. Absolute uncertainty is expressed in:

(a) the same unit as the quantity **(b)** percent **(c)** inverse unit **(d)** no unit

17. Fractional uncertainty is:

(a) $\Delta x/x$ **(b)** $\Delta x + x$ **(c)** $\Delta x \times x$ **(d)** $x/\Delta x$

18. Percentage uncertainty is:

(a) least count × 100 **(b)** fractional uncertainty × 100 **(c)** mean × 100 **(d)** absolute uncertainty × 100

19. A more precise instrument has:

(a) more mass **(b)** larger least count **(c)** larger zero error **(d)** smaller least count

20. Accuracy is best judged by:

(a) percentage uncertainty **(b)** unit used **(c)** number of digits **(d)** least count only

21. Precision refers to:

(a) closeness of repeated readings **(b)** zero error **(c)** closeness to true value **(d)** calibration

22. Accuracy refers to:

(a) closeness to true value **(b)** range **(c)** least count **(d)** closeness of repeated readings

23. The number of significant figures in 0.0025 is:

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

24. The number of significant figures in 2.50×10^3 is:

(a) 5 **(b)** 2 **(c)** 4 **(d)** 3

25. The number of significant figures in 200.0 is:

(a) 1 **(b)** 3 **(c)** 2 **(d)** 4

26. In multiplication, the answer keeps:

(a) most significant figures **(b)** least significant figures **(c)** all digits **(d)** least decimal places

27. In addition, the answer keeps:

(a) all digits **(b)** least significant figures **(c)** most decimal places **(d)** least decimal places

28. 43.75 rounded to three significant figures by the even-digit rule is:

(a) 43.7 **(b)** 44.0 **(c)** 43.8 **(d)** 43.75

29. 73.650 rounded to three significant figures by the even-digit rule is:

(a) 73.65 **(b)** 74.0 **(c)** 73.7 **(d)** 73.6

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30. 2.8723 × 1.6 to correct significant figures is:

(a) 4.596 **(b)** 4.59568 **(c)** 4.60 **(d)** 4.6

31. 12 kg + 2.02 kg + 5.1 kg to correct precision is:

(a) 19.1 kg **(b)** 19 kg **(c)** 19.0 kg **(d)** 19.12 kg

32. The dimensions of force are:

(a) $[MLT^{-2}]$ **(b)** $[ML^2T^{-2}]$ **(c)** $[ML^{-1}T^{-2}]$ **(d)** $[MLT^{-1}]$

33. The dimensions of energy are:

(a) $[ML^{-1}T^{-2}]$ **(b)** $[ML^2T^{-2}]$ **(c)** $[MLT^{-2}]$ **(d)** $[ML^2T^{-3}]$

34. The dimensions of power are:

(a) $[MLT^{-3}]$ **(b)** $[ML^2T^{-1}]$ **(c)** $[ML^2T^{-3}]$ **(d)** $[ML^2T^{-2}]$

35. The dimensions of pressure are:

(a) $[ML^2T^{-2}]$ **(b)** $[MLT^{-2}]$ **(c)** $[ML^{-2}T^{-2}]$ **(d)** $[ML^{-1}T^{-2}]$

36. The dimensions of Planck's constant are:

(a) $[ML^2T^{-3}]$ **(b)** $[ML^2T^{-1}]$ **(c)** $[MLT^{-1}]$ **(d)** $[ML^2T^{-2}]$

37. The dimensions of angular velocity are:

(a) $[MLT^{-1}]$ **(b)** $[LT^{-1}]$ **(c)** dimensionless **(d)** $[T^{-1}]$

38. The ratio of dimensions of force and energy is:

(a) $[L]$ **(b)** $[L^{-1}]$ **(c)** $[T^{-1}]$ **(d)** $[T]$

39. Which pair does NOT have identical dimensions?

(a) work and torque **(b)** impulse and momentum **(c)** moment of inertia and moment of force **(d)** angular momentum and Planck constant

40. Dimensional analysis cannot determine:

(a) dimensionless constants **(b)** dependence on base quantities **(c)** homogeneity **(d)** units

41. A dimensionless quantity:

(a) has no physical meaning **(b)** has no unit ever **(c)** may still have a unit **(d)** is always zero

42. The radian is:

(a) $[L]$ **(b)** dimensionless **(c)** $[ML]$ **(d)** $[L^{-1}]$

43. In addition or subtraction, uncertainties are combined by adding:

(a) squares **(b)** absolute uncertainties **(c)** percentage uncertainties **(d)** fractional uncertainties

44. In multiplication or division, uncertainties are combined by adding:

(a) least counts **(b)** absolute uncertainties **(c)** zero errors **(d)** percentage uncertainties

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45. If $A = \pi r^2$, the percentage uncertainty in A is:

(a) half percentage uncertainty in r **(b)** 2 × percentage uncertainty in r **(c)** 4 × percentage uncertainty in r **(d)** percentage uncertainty in r

46. If V = 3.4 ± 0.1 V and I = 0.68 ± 0.05 A, percentage uncertainty in R = V/I is about:

(a) 7% **(b)** 3% **(c)** 10% **(d)** 4%

47. A zero error is:

(a) no error **(b)** personal error only **(c)** random error **(d)** systematic error

48. Random errors can be reduced by:

(a) using a zero error **(b)** ignoring the last digit **(c)** using a larger unit **(d)** taking repeated readings and averaging

49. Scientific notation is useful because it:

(a) shows significant figures clearly **(b)** removes units **(c)** increases uncertainty **(d)** changes the value

50. The correct SI writing is:

(a) 5.0 M **(b)** 5.0 m. **(c)** 5.0 ms **(d)** 5.0 m

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Answer Key with Brief Reasons

1. Answer: (c) 7 — SI has seven base units.

2. Answer: (a) candela — Candela is the base unit of luminous intensity.

3. Answer: (b) mole — Mole measures amount of substance.

4. Answer: (a) newton — Newton = kg m s^{-2} is derived.

5. Answer: (d) pascal — Pressure = force/area = N m^{-2} = Pa.

6. Answer: (c) 10^{-9} m — Nano means 10^{-9}.

7. Answer: (b) 10^{6} — Mega means 10^{6}.

8. Answer: (b) 10^{-6} F — Micro means 10^{-6}.

9. Answer: (a) not allowed — SI allows only one prefix per unit.

10. Answer: (d) 10^{-6} m^3 — $(10^{-2}\ m)^3 = 10^{-6}\ m^3$.

11. Answer: (a) 0.1 cm — A metre rule is graduated in millimetres.

12. Answer: (b) 0.01 cm — School vernier callipers read to 0.01 cm.

13. Answer: (b) 0.001 cm — Screw gauge reads to 0.01 mm = 0.001 cm.

14. Answer: (c) least count — The smallest division limits a single reading.

15. Answer: (b) mean deviation — Mean deviation measures spread about the average.

16. Answer: (a) the same unit as the quantity — Absolute uncertainty has the unit of the measurement.

17. Answer: (a) Δx/x — Fractional uncertainty = absolute uncertainty / measured value.

18. Answer: (b) fractional uncertainty × 100 — Percentage = fractional × 100.

19. Answer: (d) smaller least count — Precision improves with smaller least count.

20. Answer: (a) percentage uncertainty — Smaller percentage uncertainty means greater accuracy.

21. Answer: (a) closeness of repeated readings — Precision is repeatability.

22. Answer: (a) closeness to true value — Accuracy is correctness.

23. Answer: (a) 2 — Leading zeros are not significant.

24. Answer: (d) 3 — Scientific notation shows three measured digits.

25. Answer: (d) 4 — Trailing zero after decimal is significant.

26. Answer: (b) least significant figures — Product is limited by the least precise factor.

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27. Answer: (d) least decimal places — Sum is limited by the least precise decimal place.

28. Answer: (c) 43.8 — The kept digit 7 becomes even 8.

29. Answer: (d) 73.6 — The kept digit 6 is already even.

30. Answer: (d) 4.6 — 1.6 has two significant figures.

31. Answer: (b) 19 kg — 12 kg has no decimal place.

32. Answer: (a) $[MLT^{-2}]$ — F = ma gives $[M][LT^{-2}]$.

33. Answer: (b) $[ML^2T^{-2}]$ — Work = force × distance.

34. Answer: (c) $[ML^2T^{-3}]$ — Power = work/time.

35. Answer: (d) $[ML^{-1}T^{-2}]$ — Pressure = force/area.

36. Answer: (b) $[ML^2T^{-1}]$ — E = hf gives h = E/f.

37. Answer: (d) $[T^{-1}]$ — Angular velocity = angle/time; angle is dimensionless.

38. Answer: (b) $[L^{-1}]$ — Force/energy = $[MLT^{-2}]/[ML^2T^{-2}] = [L^{-1}]$.

39. Answer: (c) moment of inertia and moment of force — Moment of inertia is $[ML^2]$, torque is $[ML^2T^{-2}]$.

40. Answer: (a) dimensionless constants — Constants like ½ or 2π are not found by dimensions.

41. Answer: (c) may still have a unit — Angle has unit radian but no dimensions.

42. Answer: (b) dimensionless — Radian = arc/radius.

43. Answer: (b) absolute uncertainties — Worst-case absolute uncertainties add.

44. Answer: (d) percentage uncertainties — Relative uncertainties add for products/quotients.

45. Answer: (b) 2 × percentage uncertainty in r — Power factor multiplies percentage uncertainty.

46. Answer: (c) 10% — 3% + 7% = 10%.

47. Answer: (d) systematic error — Zero error shifts all readings equally.

48. Answer: (d) taking repeated readings and averaging — Averaging reduces random scatter.

49. Answer: (a) shows significant figures clearly — It displays measured digits unambiguously.

50. Answer: (d) 5.0 m — Unit symbols are not pluralised or followed by a full stop.

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